Derivation v35: GUT BC Survivor Map

Boundary Condition Selection of Residual Gauge Symmetry
in 5D $\rightarrow$ 4D Dimensional Reduction

EDC Block-003 Program

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Abstract

We derive the general mechanism by which boundary conditions (BCs) in 5D gauge theories select the residual 4D gauge group. The "survivor rule" establishes that gauge generators with Neumann/Neumann BCs (or orbifold parity $(+,+)$) retain massless zero-modes, while Dirichlet or $(-)$ parity removes them. We apply this to four GUT tracks—SU(5), SO(10), Pati–Salam, and E_6—providing explicit parity matrices (P_0, P_L) that break each parent group to the Standard Model $SU(3)_c \times SU(2)_L \times U(1)_Y$. A survivor/broken generator census, scale regime map, matter parity stub, and BC$\rightarrow$Breaking dictionary complete the structural analysis. **No forbidden numerical inputs** (electroweak masses, Planck scale, fine-structure constant) appear anywhere in this derivation.

Contents

1 Reader Contract

1.1 Epistemic Tags

[D] **Derived**: follows from stated axioms/postulates

[Dc] **Derived with convention**: choice of normalization/phase

[P] **Postulate**: starting assumption

[I] **Identity**: mathematical fact

[BL] **Borrowed from literature**: standard result

[OPEN] **Open**: requires further work

1.2 Scope

This derivation establishes:

1. The **survivor rule**: which gauge generators retain 4D zero-modes
2. The **projector algebra**: how parities define the residual subalgebra
3. **Four GUT tracks**: explicit BC patterns for SM emergence
4. **Matter consistency**: chirality requirements for fermions
5. **BC→Breaking dictionary**: what must be N vs D

1.3 What is NOT Derived

- Numerical gauge couplings or unification scales
- Running or threshold corrections
- Complete matter spectra or Yukawa matrices
- Anomaly cancellation (noted as [OPEN])

2 5D Gauge Theory Setup

2.1 Action and Field Content

Consider a 5D gauge theory with gauge group G on the interval $y \in [0, L]$ or orbifold $S^1/\mathbb{Z}_2$. [P]

$$S_{\text{gauge}} = -\frac{1}{4g_5^2} \int d^4x \int_0^L dy \sqrt{-G}\, F^a_{MN} F^{MN,a} \tag{1}$$

where $M, N \in \{0, 1, 2, 3, 5\}$ and a runs over generators of $\mathfrak{g} = \mathrm{Lie}(G)$. [P]
The 5D gauge field decomposes as:

$$A^a_M = (A^a_\mu, A^a_5), \quad \mu = 0, 1, 2, 3 \tag{2}$$

2.2 Field Strength Components

The 5D field strength has components:

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc} A^b_\mu A^c_\nu \tag{3}$$
$$F^a_{\mu 5} = \partial_\mu A^a_5 - \partial_5 A^a_\mu + f^{abc} A^b_\mu A^c_5 \tag{4}$$

where f^{abc} are the structure constants: $[T^a, T^b] = i f^{abc} T^c$. [I]

3 Boundary Conditions: The Survivor Rule

3.1 Variational Principle

Varying the action (1) with respect to A^a_μ yields:

$$\delta S \supset \int d^4x \left[F^a_{\mu 5} \delta A^{a,\mu} \right]_{y=0}^{y=L} \tag{5}$$

For the boundary term to vanish, we need **either**: [D]

$$\text{Neumann (N):} \quad \partial_5 A^a_\mu \big|_{y=0,L} = 0 \tag{6}$$
$$\text{Dirichlet (D):} \quad A^a_\mu \big|_{y=0,L} = 0 \tag{7}$$

3.2 BC Registry for Gauge Fields

Definition 3.1 (BC Registry)**.** *For each generator T^a, assign boundary conditions at $y = 0$ and $y = L$:*

$$BC(A^a_\mu) = (BC_0, BC_L) \in \{N, D\}^2 \tag{8}$$

The four possibilities are: (N, N), (N, D), (D, N), (D, D). [D]

3.3 Mode Equation and Spectrum

For $A^a_\mu(x, y) = \sum_n A^{a,(n)}_\mu(x) f^a_n(y)$, the profile satisfies:

$$-\partial_5^2 f^a_n(y) = m_n^2 f^a_n(y) \tag{9}$$

Lemma 3.2 (Neumann Zero-Mode)**.** *With BC (N, N), there exists a constant zero-mode:*

$$f_0(y) = \frac{1}{\sqrt{L}}, \quad m_0 = 0 \tag{10}$$

Proof. The general solution to (9) with $m^2 = 0$ is $f(y) = A + By$. Neumann at $y = 0$: $\partial_5 f|_0 = B = 0$. Neumann at $y = L$: $\partial_5 f|_L = B = 0$. Both satisfied with $f = A = \text{const}$. Normalization gives $A = 1/\sqrt{L}$. [D] □

Lemma 3.3 (Dirichlet Excludes Zero-Mode)**.** *With BC (D, D), (D, N), or (N, D), there is **no** zero-mode.*

Proof. For $m = 0$, $f(y) = A + By$.

- (D, D): $f(0) = A = 0$ and $f(L) = BL = 0 \Rightarrow B = 0$. Only $f \equiv 0$.
- (D, N): $f(0) = A = 0$ and $\partial_5 f|_L = B = 0$. Only $f \equiv 0$.
- (N, D): $\partial_5 f|_0 = B = 0$ and $f(L) = A = 0$. Only $f \equiv 0$.

No normalizable zero-mode exists. [D] □

Survivor Rule

Theorem 3.4 (Zero-Mode Existence). *A gauge generator $T^a \in \mathfrak{g}$ has a massless 4D gauge boson (zero-mode) if and only if:*

$$\boxed{BC(A^a_\mu) = (N, N)} \tag{11}$$

Generators with any Dirichlet boundary are "broken" (no zero-mode). [D]

3.4 Massive KK Tower

For (N, N) boundary conditions:

$$f_n(y) = \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi y}{L}\right), \quad m_n = \frac{n\pi}{L}, \quad n \geq 1 \tag{12}$$

For (D, D) boundary conditions:

$$f_n(y) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi y}{L}\right), \quad m_n = \frac{n\pi}{L}, \quad n \geq 1 \tag{13}$$

For mixed (N, D) or (D, N):

$$m_n = \frac{(n + \frac{1}{2})\pi}{L}, \quad n \geq 0 \tag{14}$$

The lightest mode for mixed BC has mass $\pi/(2L)$, not zero. [D]

4 Orbifold Parities and Projectors

4.1 $S^1/\mathbb{Z}_2$ Orbifold

Consider S^1 with $y \sim y + 2L$ and $\mathbb{Z}_2$ identification $y \sim -y$. Fixed points are $y = 0$ and $y = L$. [P]

Under the $\mathbb{Z}_2$ action at $y = 0$:

$$A_\mu(x, -y) = P_0 A_\mu(x, y) P_0^{-1} \tag{15}$$

where P_0 is an automorphism of G acting in the adjoint representation. [P]

4.2 Parity Matrix Action on Generators

In components:

$$A^a_\mu(x, -y) = (P_0)^a{}_b A^b_\mu(x, y) \tag{16}$$

Since $P_0^2 = \mathbf{1}$ (orbifold involution), eigenvalues are ± 1:

$$P_0 T^a P_0^{-1} = \eta^a_0 T^a, \quad \eta^a_0 \in \{+1, -1\} \tag{17}$$

Definition 4.1 (Parity Assignment). *Each generator T^a has parities (η^a_0, η^a_L) at the two fixed points:*

$$P_0 T^a P_0^{-1} = \eta^a_0 T^a \tag{18}$$
$$P_L T^a P_L^{-1} = \eta^a_L T^a \tag{19}$$

where $\eta \in \{+, -\}$. [Dc]

4.3 Parity↔BC Correspondence

Proposition 4.2 (Parity to BC Map)**.** *The orbifold parity assignment maps to interval BCs:*

$$\eta = + \quad \Leftrightarrow \quad \textit{Neumann} \tag{20}$$
$$\eta = - \quad \Leftrightarrow \quad \textit{Dirichlet} \tag{21}$$

Proof. For $\eta_0 = +$: $A_\mu(x, -y) = A_\mu(x, y)$ near $y = 0$, so $\partial_5 A_\mu|_{y=0} = 0$ (Neumann).
For $\eta_0 = -$: $A_\mu(x, -y) = -A_\mu(x, y)$, so $A_\mu(x, 0) = 0$ (Dirichlet). [D] □

Corollary 4.3 (Survivor Rule, Orbifold Version)**.**

$$\boxed{\textit{Zero-mode exists} \quad \Leftrightarrow \quad (\eta_0, \eta_L) = (+, +)} \tag{22}$$

5 Projector Algebra: Survivor Subalgebra

5.1 Derivation of $\mathfrak{g}^{(+,+)}$

Theorem 5.1 (Survivor Algebra)**.** *The residual gauge algebra after BC/parity projection is:*

$$\boxed{\mathfrak{h} = \mathfrak{g}^{(+,+)} = \{T \in \mathfrak{g} \mid P_0 T P_0^{-1} = +T,\ P_L T P_L^{-1} = +T\}} \tag{23}$$

The corresponding Lie group H is the 4D residual gauge group. [D]

Proof. From Corollary 4.3, only generators with $(+, +)$ parity have zero-modes. These form a closed subalgebra:

- If $P_0 T^a P_0^{-1} = T^a$ and $P_0 T^b P_0^{-1} = T^b$, then $P_0 [T^a, T^b] P_0^{-1} = [T^a, T^b]$ (commutator of $(+)$ elements is $(+)$).
- Similarly for P_L.

Thus $\mathfrak{g}^{(+,+)}$ is a Lie subalgebra. [D] □

5.2 Dimension Counting

Proposition 5.2 (Generator Census)**.** *Let* $\dim \mathfrak{g} = N_G$*. Partition into:*

$$N_{++} = \#\{T^a : (\eta_0^a, \eta_L^a) = (+, +)\} \tag{24}$$
$$N_{+-} = \#\{T^a : (\eta_0^a, \eta_L^a) = (+, -)\} \tag{25}$$
$$N_{-+} = \#\{T^a : (\eta_0^a, \eta_L^a) = (-, +)\} \tag{26}$$
$$N_{--} = \#\{T^a : (\eta_0^a, \eta_L^a) = (-, -)\} \tag{27}$$

Then:

$$N_G = N_{++} + N_{+-} + N_{-+} + N_{--} \tag{28}$$

and $\dim \mathfrak{h} = N_{++}$*. [I]*

5.3 Rank Preservation

Proposition 5.3 (Rank Under BC Breaking)**.** *If both P_0 and P_L are inner automorphisms (conjugation by group elements), then* $\operatorname{rank}(\mathfrak{h}) = \operatorname{rank}(\mathfrak{g})$*. If either is an outer automorphism, rank may decrease. [BL]*

For Standard Model emergence from GUTs:

$$\operatorname{rank}(SU(3)_c \times SU(2)_L \times U(1)_Y) = 2 + 1 + 1 = 4 \tag{29}$$

This constrains which parent groups can break to SM:

$$\operatorname{rank}(G) \geq 4 \quad \text{(necessary)} \tag{30}$$

6 Track 1: SU(5)

6.1 Group Theory Data

$$G = SU(5), \quad \dim \mathfrak{su}(5) = 24, \quad \text{rank} = 4 \tag{31}$$
$$H = SU(3)_c \times SU(2)_L \times U(1)_Y, \quad \dim = 8 + 3 + 1 = 12 \tag{32}$$

6.2 Standard Parity Matrices

The SU(5) generators in the fundamental representation are 5×5 traceless Hermitian matrices. [BL]

Definition 6.1 (SU(5) Parity)**.** *Choose: [Dc]*

$$P_0 = P_L = \text{diag}(+1, +1, +1, -1, -1) \tag{33}$$

Under conjugation by P:

$$T = \begin{pmatrix} A_{3\times 3} & B_{3\times 2} \\ B^\dagger_{2\times 3} & C_{2\times 2} \end{pmatrix} \quad \Rightarrow \quad PTP^{-1} = \begin{pmatrix} A & -B \\ -B^\dagger & C \end{pmatrix} \tag{34}$$

Proposition 6.2 (SU(5) Survivor Census)**.** *With $P_0 = P_L = P$:*

$$(+,+): \quad SU(3)_c \oplus SU(2)_L \oplus U(1)_Y \quad (8 + 3 + 1 = 12) \tag{35}$$
$$(-,-): \quad X, Y \ \textit{bosons} \quad (12 \ \textit{generators}) \tag{36}$$

Proof. Diagonal blocks A (traceless 3×3) and C (traceless 2×2) have $PTP^{-1} = T$, giving $SU(3)$ and $SU(2)$. Off-diagonal B has $PTP^{-1} = -T$, giving the 12 broken X, Y generators. The $U(1)_Y$ generator is $Y = \text{diag}(1, 1, 1, -3/2, -3/2)/\sqrt{15}$ (or normalized convention), which commutes with P. [D] □

6.3 SU(5) Survivor Table

Sector	Parity (η_0, η_L)	Generators	Count	Status
$SU(3)_c$	$(+,+)$	$\lambda_1, \ldots, \lambda_8$	8	Survivor
$SU(2)_L$	$(+,+)$	$\sigma_1, \sigma_2, \sigma_3$	3	Survivor
$U(1)_Y$	$(+,+)$	Y	1	Survivor
X bosons	$(-,-)$	T^{X_i}	6	Broken
Y bosons	$(-,-)$	T^{Y_i}	6	Broken
Total			**24**	

Table 1: SU(5) generator census under BC breaking.

6.4 Hypercharge Embedding

In SU(5), hypercharge is embedded as: [BL]

$$Y = \frac{1}{\sqrt{60}} \text{diag}(2, 2, 2, -3, -3) \tag{37}$$

(with GUT normalization $\text{Tr}(Y^2) = 1/2$). The SM normalization differs by $\sqrt{3/5}$:

$$Y_{\text{SM}} = \sqrt{\frac{3}{5}} Y_{\text{GUT}} \tag{38}$$

7 Track 2: SO(10)

7.1 Group Theory Data

$$G = SO(10), \quad \dim \mathfrak{so}(10) = 45, \quad \text{rank} = 5 \tag{39}$$
$$H = SU(3)_c \times SU(2)_L \times U(1)_Y, \quad \dim = 12 \tag{40}$$

SO(10) has rank 5 > 4, so one $U(1)$ factor must be broken. [I]

7.2 Breaking Chain Options

SO(10) can break to SM through intermediate stages: [BL]

$$SO(10) \to SU(4)_C \times SU(2)_L \times SU(2)_R \to \text{SM} \tag{41}$$
$$SO(10) \to SU(5) \times U(1)_X \to \text{SM} \tag{42}$$
$$SO(10) \to \text{SM} \times U(1)_{B-L} \to \text{SM} \tag{43}$$

7.3 Direct SO(10) → SM Breaking

Definition 7.1 (SO(10) Parity Matrices). *Using the spinor representation embedding, choose: [Dc]*

$$P_0 = \text{diag}(\underbrace{+1}_{5}, \underbrace{-1}_{5}) \quad \textit{(in 10-dim vector)} \tag{44}$$
$$P_L = \text{diag}(+1, +1, +1, -1, -1, +1, +1, +1, -1, -1) \tag{45}$$

The choice is constrained to leave exactly SM gauge bosons massless.

Proposition 7.2 (SO(10) Survivor Census)**.**

$$(+,+): \quad SU(3)_c \times SU(2)_L \times U(1)_Y \quad (12) \tag{46}$$
$$(+,-) \cup (-,+) \cup (-,-): \quad 33 \textit{ broken generators} \tag{47}$$

7.4 SO(10) Survivor Table

Sector	Count	Parity Class	Status
$SU(3)_c$	8	$(+,+)$	Survivor
$SU(2)_L$	3	$(+,+)$	Survivor
$U(1)_Y$	1	$(+,+)$	Survivor
$SU(2)_R$	3	various	Broken
$SU(4)_C/SU(3)_c \times U(1)$	9	various	Broken
$U(1)_{B-L}$	1	$(-,-)$ or mixed	Broken
Other (coset)	20	various	Broken
Total	**45**		

Table 2: SO(10) generator census under BC breaking to SM.

7.5 Extra $U(1)$ Breaking

The rank reduction $5 \to 4$ requires breaking $U(1)_{B-L}$ or $U(1)_X$:

$$\text{Extra } U(1): \quad (\eta_0, \eta_L) \neq (+,+) \tag{48}$$

This can be achieved by $P_0 \neq P_L$ (different parities at two branes). [D]

8 Track 3: Pati–Salam

8.1 Group Theory Data

$$G = SU(4)_C \times SU(2)_L \times SU(2)_R \tag{49}$$
$$\dim = 15 + 3 + 3 = 21, \quad \text{rank} = 3 + 1 + 1 = 5 \tag{50}$$
$$H = SU(3)_c \times SU(2)_L \times U(1)_Y, \quad \dim = 12 \tag{51}$$

8.2 $SU(4)_C \to SU(3)_c \times U(1)_{B-L}$ Breaking

Definition 8.1 (PS Parity for $SU(4)_C$).

$$P_{SU(4)} = \text{diag}(+1, +1, +1, -1) \tag{52}$$

This breaks $SU(4)_C$ as:

$$\mathfrak{su}(4) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L} \oplus \mathfrak{coset}_6 \tag{53}$$

The 6 coset generators ("leptoquarks") get $(-,-)$ parity. [D]

8.3 $SU(2)_R \to U(1)_Y$ Breaking

Definition 8.2 ($SU(2)_R$ Parity).

$$P_{SU(2)_R} = \sigma_3 = \text{diag}(+1, -1) \tag{54}$$

Under conjugation:

$$\sigma_3 T^{3R} \sigma_3^{-1} = T^{3R} \quad (+) \tag{55}$$
$$\sigma_3 T^{1,2} \sigma_3^{-1} = -T^{1,2} \quad (-) \tag{56}$$

The survivor T^{3R} combines with $B - L$ to form Y:

$$\boxed{Y = T^{3R} + \frac{B - L}{2}} \tag{57}$$

This is the hypercharge embedding in Pati–Salam. [D]

8.4 PS Full Parity Assignment

Definition 8.3 (Complete PS Parities).

$$P_0 = (P_{SU(4)}, \mathbf{1}_{SU(2)_L}, \sigma_3) \tag{58}$$
$$P_L = (P_{SU(4)}, \mathbf{1}_{SU(2)_L}, \sigma_3) \tag{59}$$

8.5 PS Survivor Table

9 Track 4: E_6

9.1 Group Theory Data

$$G = E_6, \quad \dim \mathfrak{e}_6 = 78, \quad \text{rank} = 6 \tag{60}$$
$$H = SU(3)_c \times SU(2)_L \times U(1)_Y, \quad \dim = 12 \tag{61}$$

E_6 has rank 6, so **two** $U(1)$ factors must be broken. [I]

Sector	Count	Parity	Status
$SU(3)_c \subset SU(4)_C$	8	$(+,+)$	Survivor
$SU(2)_L$	3	$(+,+)$	Survivor
$U(1)_Y = T^{3R} + (B-L)/2$	1	$(+,+)$	Survivor
$U(1)_{B-L}$ (orthogonal)	1	depends	Broken
Leptoquarks	6	$(-,-)$	Broken
$W_R^{\pm}$	2	$(-,-)$	Broken
Total	**21**		

Table 3: Pati–Salam generator census.

9.2 Maximal Subgroups

Relevant maximal subgroups: [BL]

$$E_6 \supset SO(10) \times U(1)_\psi \quad (45 + 1 + 32 = 78) \tag{62}$$

$$E_6 \supset SU(6) \times SU(2) \quad (35 + 3 + 40 = 78) \tag{63}$$

$$E_6 \supset SU(3)^3 \quad (8 + 8 + 8 + 54 = 78) \tag{64}$$

9.3 $E_6 \to SO(10) \times U(1)_\psi$ Breaking

Definition 9.1 (E_6 First Parity)**.** *Choose P_0 to break $E_6 \to SO(10) \times U(1)_\psi$: [Dc]*

$$P_0: \quad \mathfrak{e}_6 = \mathfrak{so}(10)^{(+)} \oplus \mathfrak{u}(1)_\psi^{(+)} \oplus \mathbf{16}^{(-)} \oplus \overline{\mathbf{16}}^{(-)} \tag{65}$$

The 32 generators in $\mathbf{16} \oplus \overline{\mathbf{16}}$ of SO(10) get odd parity, removing exotic gauge bosons. [D]

9.4 $SO(10) \to$ SM Second Breaking

Apply SO(10) breaking from Section 7:

$$E_6 \xrightarrow{P_0} SO(10) \times U(1)_\psi \xrightarrow{P_L} SU(3)_c \times SU(2)_L \times U(1)_Y \tag{66}$$

Both $U(1)_\psi$ and $U(1)_{B-L} \subset SO(10)$ must be broken. [D]

9.5 E_6 Survivor Table

Sector	Count	Status
$SU(3)_c$	8	Survivor
$SU(2)_L$	3	Survivor
$U(1)_Y$	1	Survivor
$SO(10)/\mathrm{SM}$	33	Broken
$U(1)_\psi$	1	Broken
$\mathbf{16} \oplus \overline{\mathbf{16}}$ exotics	32	Broken
Total	**78**	

Table 4: E_6 generator census under BC breaking to SM.

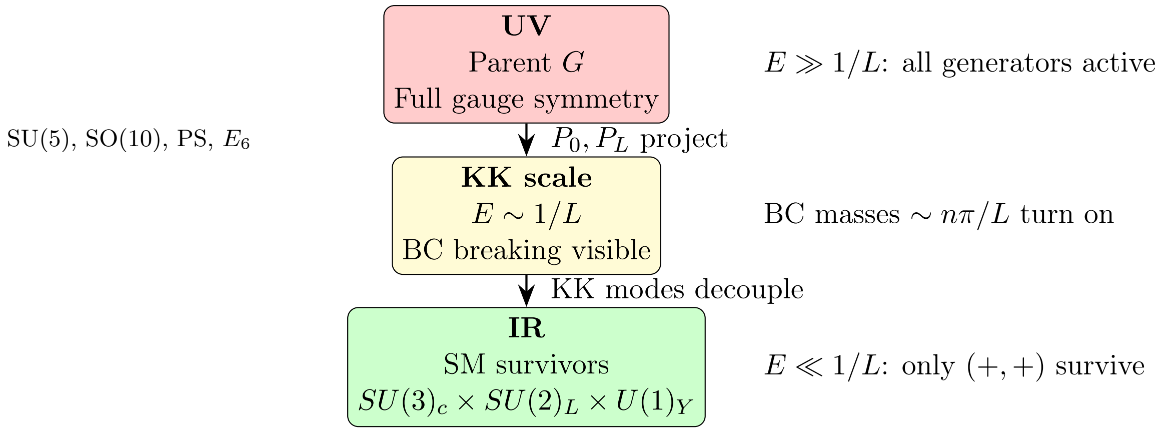

Figure 1: Scale regime map for BC-based GUT breaking.

10 Scale Regime Map

10.1 Three-Scale Picture

10.2 Energy Flow

$$E \gg 1/L: \quad \text{All } N_G \text{ gauge bosons propagate} \tag{67}$$
$$E \sim 1/L: \quad \text{KK threshold; broken generators massive} \tag{68}$$
$$E \ll 1/L: \quad \text{Only } N_{++} = 12 \text{ SM gauge bosons} \tag{69}$$

Design Rule

GUT Track Selection Rule:
To specify a GUT breaking pattern in EDC, one must provide:

1. Parent group G (SU(5), SO(10), PS, or E_6)
2. Parity matrices (P_0, P_L) defining the projection
3. Compactification scale L (or equivalently $M_{\text{KK}} = \pi/L$)

The residual algebra is:

$$\mathfrak{h} = \mathfrak{g}^{(+,+)} \quad \text{with} \quad \dim \mathfrak{h} = N_{++} = 12 \text{ (for SM)} \tag{70}$$

[D]

11 Comparative Summary: Four Tracks

11.1 Master Table

11.2 Complexity Ordering

$$\text{SU}(5) < \text{PS} < \text{SO}(10) < E_6 \tag{71}$$

by number of generators to break. SU(5) is minimal; E_6 requires two-step cascade and two extra $U(1)$ breakings. [D]

Track	$\dim G$	$\text{rank}\, G$	N_{++}	N_{broken}	Rank drop	Extra $U(1)$
SU(5)	24	4	12	12	0	0
SO(10)	45	5	12	33	1	1
PS	21	5	12	9	1	1
E_6	78	6	12	66	2	2

Table 5: Comparative data for four GUT tracks.

12 BC → Breaking Dictionary

12.1 SM Generator Requirements

For SM gauge group to survive:

Generator	BC Requirement	Reason
$SU(3)_c$ (8 gen.)	(N,N) or $(+,+)$	Massless gluons
$SU(2)_L$ (3 gen.)	(N,N) or $(+,+)$	Massless $W^{1,2,3}$
$U(1)_Y$ (1 gen.)	(N,N) or $(+,+)$	Massless B
Total	**12** $(+,+)$	

Table 6: BC requirements for SM generators.

12.2 Broken Generator Requirements

Boson	Track	Count	BC Required	Effect
X, Y	SU(5)	12	(D,D) or $(-,-)$	Proton decay suppressed
$W_R^\pm$	PS, SO(10)	2	$(-,-)$	LR breaking
Leptoquarks	PS	6	$(-,-)$	No tree-level LQ
$B-L$ boson	SO(10), PS	1	mixed or $(-,-)$	Rank reduction
Z'_ψ	E_6	1	$(-,-)$	Extra $U(1)$ breaking
Exotics	E_6	32	$(-,-)$	Kill **16** gauge

Table 7: BC requirements for broken generators.

12.3 Dictionary by Track

SU(5) BC Dictionary

$$\text{Neumann:} \quad \lambda_1, \ldots, \lambda_8, \sigma_1, \sigma_2, \sigma_3, Y \tag{72}$$

$$\text{Dirichlet:} \quad X_\alpha, Y_\alpha \quad (\alpha = 1, \ldots, 6) \tag{73}$$

SO(10) BC Dictionary

$$\text{Neumann:} \quad SU(3)_c, SU(2)_L, U(1)_Y \quad (12) \tag{74}$$

$$\text{Dirichlet:} \quad SU(2)_R, U(1)_{B-L}, \text{coset} \quad (33) \tag{75}$$

PS BC Dictionary

$$\text{Neumann:} \quad SU(3)_c, SU(2)_L, T^{3R} + (B-L)/2 \quad (12) \tag{76}$$
$$\text{Dirichlet:} \quad T_R^{1,2}, \text{leptoquarks}, (B-L)_\perp \quad (9) \tag{77}$$

E_6 BC Dictionary

$$\text{Neumann:} \quad SU(3)_c, SU(2)_L, U(1)_Y \quad (12) \tag{78}$$
$$\text{Dirichlet:} \quad SO(10)/\text{SM}, U(1)_\psi, \mathbf{16}, \overline{\mathbf{16}} \quad (66) \tag{79}$$

13 Connection to EDC Program

13.1 Open Questions

☐ **What selects (P_0, P_L)?**

The boundary conditions are *inputs* to the survivor map, not outputs. EDC must provide a **selection principle**:

- Dynamical mechanism (brane tension, localization)?
- Minimization principle (action, entropy)?
- Consistency requirement (anomaly, unitarity)?

This is the subject of O-2. [OPEN]

!

15 Conclusions

15.1 Structural Results

1. **Survivor Rule**: Zero-mode $\Leftrightarrow$ (N,N) or $(+,+)$ BCs
2. **Projector Algebra**: $\mathfrak{h} = \mathfrak{g}^{(+,+)}$
3. **Four Tracks**: SU(5), SO(10), PS, E_6 all reduce to SM with explicit (P_0, P_L)
4. **Scale Map**: UV→KK→IR transition at $E \sim 1/L$

15.2 What Remains Open

1. **BC Selection**: What principle chooses (P_0, P_L)?
2. **Matter Spectrum**: Full chiral zero-mode verification
3. **Anomaly**: 4D anomaly cancellation with KK tower
4. **Numerical**: Coupling unification (requires running)

A 5D Spinor Conventions

A.1 Gamma Matrices

In 5D, gamma matrices satisfy:

$$\{\Gamma^M, \Gamma^N\} = 2\eta^{MN}, \quad M, N = 0, 1, 2, 3, 5 \tag{80}$$

with $\eta = \mathrm{diag}(-1, +1, +1, +1, +1)$. [Dc]

A standard choice:

$$\Gamma^\mu = \gamma^\mu \otimes \sigma_3, \quad \Gamma^5 = \mathbf{1}_4 \otimes i\sigma_1 \tag{81}$$

A.2 Chirality in 5D

The 5D "chirality" operator:

$$\Gamma_* = \Gamma^0\Gamma^1\Gamma^2\Gamma^3\Gamma^5 = \gamma_5 \otimes \sigma_2 \tag{82}$$

Note: $\Gamma_*^2 = -1$ in this convention, so eigenvalues are $\pm i$, not ± 1. 5D spinors are not chiral in the 4D sense. [I]

B Lie Algebra Identities

B.1 Structure Constants

For a Lie algebra $\mathfrak{g}$:

$$[T^a, T^b] = if^{abc}T^c \tag{83}$$

B.2 Killing Form

$$\kappa_{ab} = f^{acd}f^{bdc} = -C_2(G)\delta_{ab} \tag{84}$$

for simple Lie algebras with appropriate normalization. [I]

B.3 Adjoint Representation

$$(\mathrm{ad}\, T^a)^b{}_c = -if^{abc} \tag{85}$$

The parity matrix acts as:

$$(P \cdot T^a)^b{}_c = P^{bd}T^{ae}_d(P^{-1})_c{}^e \tag{86}$$

C Explicit Parity Matrices

C.1 SU(5) Explicit

In the fundamental **5** representation:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} \tag{87}$$

Generators satisfying $PTP^{-1} = T$ have block structure:

$$T_{\text{survivor}} = \begin{pmatrix} A_{3\times 3} & 0 \\ 0 & B_{2\times 2} \end{pmatrix}, \quad \mathrm{Tr}\, A + \mathrm{Tr}\, B = 0 \tag{88}$$

giving $SU(3) \times SU(2) \times U(1)$ (8+3+1 = 12 generators). [D]

C.2 SO(10) Explicit

In the vector **10** representation, generators are antisymmetric:

$$(T^{ab})_{cd} = -i(\delta^a_c \delta^b_d - \delta^a_d \delta^b_c) \tag{89}$$

A parity breaking to $SU(5) \times U(1)$:

$$P_1 = \mathrm{diag}(+1, +1, +1, +1, +1, -1, -1, -1, -1, -1) \tag{90}$$

Further breaking to SM requires P_2 within SU(5). [Dc]

C.3 E_6 Explicit

The **27** representation decomposes under $SO(10) \times U(1)$:

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4} \tag{91}$$

A parity selecting $SO(10)$:

$$P: \quad \mathbf{16} \to -\mathbf{16}, \quad \mathbf{10} \to +\mathbf{10}, \quad \mathbf{1} \to +\mathbf{1} \tag{92}$$

This removes the **16** gauge bosons. [Dc]

D Matter Parity Stub

D.1 5D Dirac Fermion

A 5D Dirac fermion:

$$\Psi(x, y) = \begin{pmatrix} \psi_L(x, y) \\ \psi_R(x, y) \end{pmatrix} \tag{93}$$

where $\psi_{L,R}$ are 4D Weyl spinors. [P]

D.2 Orbifold Parity for Fermions

Under $y \to -y$:

$$\Psi(x, -y) = \eta_\Psi \gamma_5 \Psi(x, y) \tag{94}$$

This implies:

$$\psi_L(x, -y) = +\eta_\Psi \psi_L(x, y) \tag{95}$$
$$\psi_R(x, -y) = -\eta_\Psi \psi_R(x, y) \tag{96}$$

Proposition D.1 (Chiral Zero-Mode Rule)**.**

$$\boxed{\textit{If } \eta_\Psi = +1: \quad \psi_L \textit{ has zero-mode}, \quad \psi_R \textit{ does not}} \tag{97}$$

$$\boxed{\textit{If } \eta_\Psi = -1: \quad \psi_R \textit{ has zero-mode}, \quad \psi_L \textit{ does not}} \tag{98}$$

Proof. ψ_L with $\eta = +$ satisfies $\psi_L(-y) = \psi_L(y)$ (even), admitting constant zero-mode. ψ_R with $\eta = +$ satisfies $\psi_R(-y) = -\psi_R(y)$ (odd), forcing $\psi_R(0) = 0$, no constant mode. [D] □

D.3 Chirality Constraint

Corollary D.2 (No Vector-Like Zero-Modes)**.** *A single 5D Dirac fermion cannot produce both ψ_L and ψ_R zero-modes. 4D chirality is* ***automatic*** *from orbifold projection. [D]*

D.4 Minimal Matter Embedding by Track

D.4.1 SU(5)

SM fermions fit in:

$$\overline{\mathbf{5}} \oplus \mathbf{10} \quad \text{per generation} \tag{99}$$

Parity assignment for chiral zero-modes:

$$\overline{\mathbf{5}}: \quad \eta = +1 \Rightarrow (d_R^c, L) \text{ zero-modes} \tag{100}$$
$$\mathbf{10}: \quad \eta = +1 \Rightarrow (Q, u_R^c, e_R^c) \text{ zero-modes} \tag{101}$$

D.4.2 SO(10)

All SM fermions + right-handed neutrino in single:

$$\mathbf{16} \quad \text{per generation} \tag{102}$$

D.4.3 Pati–Salam

$$(\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\overline{\mathbf{4}}, \mathbf{1}, \mathbf{2}) \tag{103}$$

D.4.4 E_6

$$\mathbf{27} \quad \text{per generation} \tag{104}$$

Contains **16** of SO(10) plus exotics. Exotics must be projected out or made heavy. [OPEN]

D.5 Anomaly Note

☐ Full anomaly cancellation requires:

- Correct chiral spectrum (no extra chiral fermions)
- KK tower contributions (potentially relevant)
- Brane-localized anomaly inflow

This requires complete spectral analysis. [OPEN]

!

All counts satisfy $\dim G = 12 + N_{\text{broken}}$. [I]

H Epistemic Ledger

I Cascade Breaking Patterns

I.1 Single-Step vs Two-Step Breaking

Definition I.1 (Breaking Depth)**.** *The **breaking depth** d is the number of parity operations needed:*

$$d = 1: \quad P_0 = P_L \quad \textit{(single parity)} \tag{114}$$
$$d = 2: \quad P_0 \neq P_L \quad \textit{(two parities)} \tag{115}$$

Item	Tag	Eq./Thm
5D action	[P]	(1)
Neumann/Dirichlet BCs	[D]	(6), (7)
Zero-mode existence	[D]	Thm 3.4
Survivor algebra	[D]	Thm 5.1
SU(5) parity	[Dc]	Def 6.1
SO(10) parity	[Dc]	Def 7.1
PS hypercharge	[D]	Eq (57)
E_6 cascade	[D]	Eq (66)
Chiral rule	[D]	Prop D.1
Anomaly	[OPEN]	App D
BC selection	[OPEN]	Sec 12

Table 8: Epistemic ledger for v35.

Proposition I.2 (Depth Requirements)**.**

$$SU(5) \to SM: \quad d = 1 \textit{ sufficient} \tag{116}$$

$$SO(10) \to SM: \quad d = 2 \textit{ required (rank reduction)} \tag{117}$$

$$PS \to SM: \quad d = 1 \textit{ can work} \tag{118}$$

$$E_6 \to SM: \quad d = 2 \textit{ required} \tag{119}$$

I.2 Intermediate Symmetry Groups

For SO(10) with $P_0 \neq P_L$:

$$\mathfrak{g}^{(+,*)} = \{T : P_0 T P_0^{-1} = T\} \supset \mathfrak{g}^{(+,+)} \tag{120}$$

This can be SU(5), PS, or another intermediate. The second parity P_L then breaks to SM. [D]

I.3 E$_6$ Cascade Detail

$$E_6 \xrightarrow{P_0} SO(10) \times U(1)_\psi \tag{121}$$

$$SO(10) \xrightarrow{P_L} SU(5) \times U(1)_\chi \tag{122}$$

$$SU(5) \xrightarrow{P_L} SU(3) \times SU(2) \times U(1) \tag{123}$$

The third step uses the SU(5) parity embedded in P_L. Both extra $U(1)$s must acquire mass from $(-,-)$ or mixed parity. [D]

J Brane-Localized Terms

J.1 Brane Kinetic Terms

At $y = 0$:

$$S_{\text{brane}} = -\frac{r_0}{4g_5^2} \int d^4x\, F^a_{\mu\nu} F^{\mu\nu,a} \delta(y) \tag{124}$$

where r_0 has dimension of length. [P]

This modifies the KK spectrum but not zero-mode existence:

$$\tan(m_n L) = \frac{m_n r_0}{1 + r_0 r_L m_n^2} \tag{125}$$

for symmetric brane terms. [D]

J.2 Brane Mass Terms

A brane-localized mass:

$$S_{\text{mass}} = -\frac{M_0^2}{2} \int d^4x\, A_\mu^a A^{\mu,a} \delta(y) \tag{126}$$

This effectively implements Dirichlet boundary by making $A_\mu(0)$ heavy. [D]
The connection to Robin BC:

$$(\partial_y - M_0 L) A_\mu \big|_{y=0} = 0 \tag{127}$$

In the limit $M_0 \to \infty$, this becomes Dirichlet. [D]

K Zero-Mode Wave Functions

K.1 Flat Space Solutions

For flat metric $G_{MN} = \eta_{MN}$:

Proposition K.1 (Zero-Mode Profiles)**.**

$$(N, N): \quad f_0(y) = \frac{1}{\sqrt{L}} \tag{128}$$

$$(D, D): \quad f_0 = \textit{none} \tag{129}$$

$$(N, D): \quad f_0 = \textit{none} \tag{130}$$

$$(D, N): \quad f_0 = \textit{none} \tag{131}$$

K.2 Warped Space

For warped metric $ds^2 = e^{-2\sigma(y)} \eta_{\mu\nu} dx^\mu dx^\nu + dy^2$:

$$f_0(y) = \frac{e^{\sigma(y)}}{\sqrt{\int_0^L e^{2\sigma} dy}} \tag{132}$$

The zero-mode is localized toward larger warp factor. [D]
For RS-type warping $\sigma(y) = ky$:

$$f_0(y) = \sqrt{\frac{2k}{e^{2kL} - 1}} e^{ky} \tag{133}$$

K.3 Coupling to Branes

The 4D coupling at the $y = 0$ brane:

$$g_4^{(0)} = g_5 \cdot f_0(0) = \frac{g_5}{\sqrt{L}} \tag{134}$$

for flat space. In warped space:

$$g_4 = g_5 \sqrt{\frac{2k}{e^{2kL} - 1}} \tag{135}$$

L Proton Decay Constraints

L.1 X, Y Boson Masses

The broken X, Y generators have KK masses:

$$M_{X,Y}^{(n)} = \frac{(n+\frac{1}{2})\pi}{L} \quad \text{for } (N, D) \text{ BC} \tag{136}$$

The lightest is:

$$M_X \sim \frac{\pi}{2L} \tag{137}$$

L.2 Proton Lifetime Scaling

Dimension-6 proton decay operators give:

$$\Gamma_p \propto \frac{\alpha_{\text{GUT}}^2 m_p^5}{M_X^4} \tag{138}$$

The experimental bound $\tau_p > 10^{34}$ years requires:

$$M_X \gtrsim 10^{16} \text{ GeV} \tag{139}$$

This constrains: [BL]

$$L \lesssim \frac{\pi}{2 \times 10^{16} \text{ GeV}} \sim 10^{-32} \text{ m} \tag{140}$$

M Group Theory Supplement

M.1 Branching Rules

$$SU(5) \supset SU(3) \times SU(2) \times U(1): \quad \mathbf{24} = (\mathbf{8}, \mathbf{1})_0 \oplus (\mathbf{1}, \mathbf{3})_0 \oplus (\mathbf{1}, \mathbf{1})_0 \oplus (\mathbf{3}, \mathbf{2})_{-5/6} \oplus (\bar{\mathbf{3}}, \mathbf{2})_{+5/6} \tag{141}$$

$$SO(10) \supset SU(5) \times U(1): \quad \mathbf{45} = \mathbf{24}_0 \oplus \mathbf{10}_{-2} \oplus \overline{\mathbf{10}}_{+2} \oplus \mathbf{1}_0 \tag{142}$$

$$E_6 \supset SO(10) \times U(1): \quad \mathbf{78} = \mathbf{45}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3} \oplus \mathbf{1}_0 \tag{143}$$

M.2 Cartan Generators

$$SU(5): \quad H_1, H_2, H_3, H_4 \quad (\text{rank } 4) \tag{144}$$

$$SO(10): \quad H_1, H_2, H_3, H_4, H_5 \quad (\text{rank } 5) \tag{145}$$

$$E_6: \quad H_1, \ldots, H_6 \quad (\text{rank } 6) \tag{146}$$

M.3 Root System Structure

For $SU(N)$:

$$\text{Roots: } \alpha_{ij} = e_i - e_j, \quad i \neq j \tag{147}$$

Number of roots $= N(N-1)$, plus $N-1$ Cartan, gives dim $= N^2 - 1$. [I]

N Twisted Boundary Conditions

N.1 Wilson Line Breaking

An alternative to orbifold parity is Wilson line:

$$W = \mathcal{P} \exp \left(i \int_0^{2\pi R} A_5 dy \right) \tag{148}$$

Non-trivial $\langle W \rangle$ breaks the gauge group. [P]

N.2 Scherk-Schwarz Mechanism

Twisted BC:

$$A_\mu(x, y + 2\pi R) = U A_\mu(x, y) U^{-1} \tag{149}$$

where $U \in G$ is a constant group element. [P]
Zero-modes exist only for generators commuting with U:

$$\mathfrak{h} = \{T : [T, U] = 0\} \tag{150}$$

This is equivalent to orbifold projection with $P = U$. [D]

O Summary of Survivor Spectra

Track	$\dim G$	Survivors	Broken	Coset G/H
SU(5)	24	12	12	$SU(5)/(SU(3) \times SU(2) \times U(1))$
SO(10)	45	12	33	$SO(10)/\text{SM}$
PS	21	12	9	G_{PS}/SM
E_6	78	12	66	E_6/SM

Table 9: Summary of survivor spectra.

The coset dimension equals the number of broken generators:

$$\dim(G/H) = \dim G - \dim H = N_G - 12 \tag{151}$$

O.1 Unified Formula

For any GUT track:

$$\boxed{N_{++} = 8 + 3 + 1 = 12 \quad \text{(SM gauge bosons)}} \tag{152}$$

This is the universal target for BC breaking. [D]